Gloss-Free Sign Language Translation: GFSLT-VLP with a Keypoint Stream

Team: Claude5 — **GitHub:** `https://github.com/username/project-repo`

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I. Abstract (Maximum 100 Words)

Sign language translation usually depends on glosses, sign-by-sign annotations that are expensive and discard facial and spatial information; gloss-free models avoid them but trail gloss-based systems by about 7 BLEU-4 on PHOENIX14T. Without glosses, nothing tells the model which visual details carry meaning. GFSLT-VLP first pretrains the visual encoder with a CLIP-style contrastive loss aligning each video with its sentence, plus masked sentence restoration, then fine-tunes an encoder–decoder translator. This injects sentence-level semantics into visual features without gloss supervision, letting a gloss-free model match early gloss-based systems (21.44 vs. 21.32 test BLEU-4).

II. Proposed Novelty & Technical Contribution

1. **Novelty:** we add a *second input stream of explicit body, hand and face keypoints* to GFSLT-VLP. MediaPipe gives 75 points per frame (11 upper-body, 2×21 hand, 22 face), normalised, embedded by a small network and fused with the ResNet18 frame features before the temporal blocks.
2. **Type:** an *architectural modification* (a new input module plus early feature fusion). Losses, decoder and schedule are unchanged; the stream is used in both VLP and translation.
3. **Difference from the baseline:** GFSLT-VLP sees only RGB frames, so finger shape and hand position must be inferred from a 224×224 image where a hand spans a few dozen pixels; our stream supplies them as explicit, signer-normalised coordinates. Keypoints come from an off-the-shelf detector, not human labels, so the method stays gloss-free. On a free single-GPU budget (Kaggle T4) we reimplemented the paper with a frozen ResNet18 on pre-extracted features, shared by all compared models.

III. Theoretical / Mathematical Justification

1. **Formulation.** Let I_t be frame t and $p_{t,j} \in \mathbb{R}^2$ keypoint j. With shoulder midpoint $c = \mathrm{med}_t \frac{1}{2}(p_{t,\mathrm{LS}} + p_{t,\mathrm{RS}})$ and width $s = \mathrm{med}_t \|p_{t,\mathrm{LS}} - p_{t,\mathrm{RS}}\|$:

$$\hat{p}_{t,j} = \frac{p_{t,j} - c}{s}, \quad k_t = \big[\hat{p}_{t,1..75};\, m_t\big] \in \mathbb{R}^{225}, \quad g(k_t) = \sigma\big(W_2\, \sigma(\mathrm{LN}(W_1 k_t + b_1)) + b_2\big) \in \mathbb{R}^{256},$$

$$x_t = \big[f_t;\, g(k_t)\big] \in \mathbb{R}^{768},\ f_t = \phi_{\mathrm{R18}}(I_t) \in \mathbb{R}^{512}, \quad Z = \mathrm{Enc}\big(\mathrm{TConv}(x_{1:T})\big), \quad \mathcal{L}_{\mathrm{SLT}} = -\textstyle\sum_u \log p_\theta(y_u \mid y_{<u}, Z).$$

2. **Variables and reasoning.** $m_t \in \{0,1\}^{75}$: detection flags (missed points interpolated in time); σ: ReLU; LN: layer norm; $W_1 \in \mathbb{R}^{256\times 225}$, $W_2 \in \mathbb{R}^{256\times 256}$; ϕ_{R18}: frozen ImageNet ResNet18; TConv: two Conv1D(k=5)+pool blocks; Enc: 3-layer transformer; VLP keeps $\mathcal{L} = \mathcal{L}_s + 0.1\,\mathcal{L}_c$. (a) *Nesting:* zeroing the first-convolution weights on the 256 keypoint channels recovers the baseline exactly, so the baseline is a special case and the best achievable loss cannot increase. (b) *Invariance:* $\hat{p}$ is unchanged under $p \mapsto ap + b$ ($a > 0$), removing signer position and camera distance as nuisance variation. (c) *Motion:* the k=5 Conv1D can compute $k_{t+1} - k_t$, i.e. hand velocity. (d) *Complementarity:* f_t is globally pooled over the frame, compressing hand configuration; k_t encodes it directly.
3. **Causal link.** (a)–(d) show the stream *can* help, not that it must. It adds only 0.78M parameters (+2.9%), so capacity alone is an unlikely explanation, and the ablation below (same seed, data and schedule; only the stream differs) isolates its effect.

IV. Comparative Results & Empirical Evaluation

1. **Setup:** RWTH-PHOENIX-Weather 2014T, split 7096/519/642, glosses never used. Frames 210×260 resized to 256 and centre-cropped to 224 for ResNet18; MediaPipe at native resolution on every 2nd frame. One T4; 60 VLP + 100 translation epochs; beam 5.
2. **Comparison:** Table 1; reproduced GFSLT and GFSLT-VLP are the baselines, "+ KP" adds our stream.
3. **Takeaway:** the keypoint stream improves test BLEU-4 by **+3.58** over reproduced GFSLT (11.48→15.06) and by **+2.74** over reproduced GFSLT-VLP (12.15→14.89), with gains on every metric and split.

Table 1: Quantitative comparison of the reproduced baselines and the proposed keypoint stream on PHOENIX14T (gloss-free). Best results are highlighted in bold. Params: trainable translation model, excluding the frozen ResNet18 (11.18M) shared by all rows. Latency: mean per test video on one T4 from pre-extracted features (visual encoder + beam-5 decoding); feature extraction is extra: ResNet18 43.5 ms per video (T4), MediaPipe ∼2.5 s per video (4 CPU cores).

Method	Dev B4 (↑)	Test B1 (↑)	Test B4 (↑)	Test ROUGE (↑)	Params (M)	Latency (ms ↓)
GFSLT, reproduced (baseline)	11.26	28.62	11.48	28.59	26.44	78.0
GFSLT-VLP, reproduced (baseline)	11.52	30.57	12.15	29.90	26.44	77.9
Ours: GFSLT + KP	**13.96**	33.99	**15.06**	33.62	27.22	76.9
Ours: GFSLT-VLP + KP	13.79	**34.43**	14.89	**33.93**	27.22	76.9
Δ (GFSLT + KP – GFSLT)	+2.70	+5.37	+3.58	+5.03	+0.78	−1.1
Δ (GFSLT-VLP + KP – GFSLT-VLP)	+2.27	+3.86	+2.74	+4.03	+0.78	−1.0
GFSLT-VLP, paper (full fine-tuning, 4 GPUs)	*22.12*	*43.71*	*21.44*	*42.49*	–	–

Result Interpretation: Keypoints dominate (+2.7–3.6 test BLEU-4, +4–5 ROUGE-L) while VLP adds +0.67 / −0.17, matching the paper's Table 3 (+0.11 without strong image augmentation, which frozen features rule out). Keypoints fix sentence type, dates and word order ("im westen ist es freundlich": baseline "schönen abend noch", ours exact); numbers remain the main errors and hands are found in only ∼60% of frames. A fine-tuned CNN partly learns hand detail, so we expect a smaller, still untested gain in the full GFSLT-VLP (keypoints also help the fine-tuned TS-SLT). Translation latency is unchanged (∼1 ms noise); the extra cost is CPU keypoint extraction. One seed per model: VLP differences are within noise; the keypoint gain is consistent everywhere.